

Datafest 2024 EDA

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```
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.0
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

knitr::opts_chunk$set(tidy.opts = list(width.cutoff = 30), tidy = TRUE)
```

Load and preprocess items.csv for analysis

```
items <- read.csv("C:/Users/maxch/Downloads/Data/Random Sample of Data Files/items.csv",
  stringsAsFactors = TRUE)
items <- select(items, page, lrn_type,
  lrn_template_name, chapter_number,
  section_number, review_flag)
items <- items[!is.na(items["chapter_number"]),
  ]

# make a new column combining
# chapter number with section
# (e.g. chapter 1 section 2
# becomes 1.2)
items["chapter_section"] <- as.double(unlist(items["chapter_number"])) +
  as.double(unlist(items["section_number"]))/10

# count the most frequently
# occurring types of items
# and filter for only the
# most common ones
table(items["lrn_type"])

## lrn_type
##      association      choicematrix      clozeassociation
##             31             62             5
##      formulaV2 imageclozeassociation      mcq
##             1             2             842
```

```
##          plaintext          shorttext          sortlist
##          209                37                1

frequent_items <- filter(items,
  lrn_type == "association" |
  lrn_type == "choicematrix" |
  lrn_type == "plaintext" |
  lrn_type == "shorttext" |
  lrn_type == "mcq")
head(frequent_items)
```

```
##          page          lrn_type          lrn_template_name
## 1 1.4 Introduction to R Functions choicematrix Choice matrix - standard
## 2      1.6 Goals of This Course  plaintext Essay with plain text
## 3 1.7 Chapter 1 Review Questions      mcq Multiple choice - block layout
## 4 1.7 Chapter 1 Review Questions      mcq Multiple choice - block layout
## 5 1.7 Chapter 1 Review Questions      mcq Multiple choice - block layout
## 6 1.7 Chapter 1 Review Questions      mcq Multiple choice - block layout
##  chapter_number section_number review_flag chapter_section
## 1              1              4      FALSE             1.4
## 2              1              6      FALSE             1.6
## 3              1              7       TRUE             1.7
## 4              1              7       TRUE             1.7
## 5              1              7       TRUE             1.7
## 6              1              7       TRUE             1.7
```

Load and preprocess media_views.csv for analysis

```
media_views <- read.csv("C:/Users/maxch/Downloads/Data/Random Sample of Data Files/media_views.csv")
media_views <- select(media_views,
  book, release, chapter_number,
  section_number, media_id, access_count,
  proportion_video, proportion_time,
  review_flag)
cat("Number of observations before omitting NA's:",
  nrow(media_views))
```

```
## Number of observations before omitting NA's: 926
```

```
media_views <- na.omit(media_views)
cat("\nNumber of observations after omitting NA's:",
  nrow(media_views))
```

```
##
## Number of observations after omitting NA's: 224
```

```
head(media_views)
```

```
##          book release chapter_number
## 13 College / Statistics and Data Science (ABC) v5.0 4
## 17 College / Statistics and Data Science (ABC) v5.0 4
## 18 College / Statistics and Data Science (ABC) v5.0 4
## 29 College / Statistics and Data Science (ABC) v5.0 5
## 42 College / Advanced Statistics and Data Science (ABCD) v5.0 4
## 49 College / Advanced Statistics and Data Science (ABCD) v5.0 4
##  section_number media_id access_count proportion_video proportion_time
```

```
## 13          2 379060892          1          0.5205001          0.5205001
## 17          2 379319375          1          0.9998323          0.9998323
## 18          2 379060892          1          0.9738690          0.9738690
## 29          4 381974697          1          0.1164375          0.1164375
## 42          2 379319375          1          0.9998323          0.9998323
## 49          2 379319375          1          0.4680822          0.4680822
##      review_flag
## 13          FALSE
## 17          FALSE
## 18          FALSE
## 29          FALSE
## 42          FALSE
## 49          FALSE
```

```
# Create summary statistics
# for proportion_video per
# video, along with the
# number of total accesses
tapply(as.numeric(unlist(media_views["proportion_video"])),
      (as.character(unlist((media_views["media_id"])))),
      function(x) {
        c(unlist(summary(x)), Num_Accesses = length(x))
      })
```

```
## $'379060892'
##      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.
## 0.002166519 0.211447783 0.937669689 0.663313288 0.999777682 0.999777686
## Num_Accesses
## 55.000000000
##
## $'379150092'
##      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.
## 0.00000000 0.2363260 0.9945476 0.6601972 0.9998543 0.9998543
## Num_Accesses
## 31.0000000
##
## $'379319375'
##      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.
## 0.00000000 0.6585951 0.9998323 0.8082114 0.9998323 1.0000000
## Num_Accesses
## 64.0000000
##
## $'379319558'
##      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.
## 0.01209533 0.48062981 0.99984142 0.73263642 0.99984142 1.00000000
## Num_Accesses
## 32.000000000
##
## $'381974697'
##      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.
## 0.005063083 0.154160714 0.691145870 0.598180644 0.999920680 1.000000000
## Num_Accesses
## 42.000000000
```

```
(sapply(media_views["proportion_video"],
summary))
```

```
##           proportion_video
## Min.          0.0000000
## 1st Qu.       0.3309119
## Median        0.9745336
## Mean          0.7019724
## 3rd Qu.       0.9998323
## Max.          1.0000000
```

Notes: 379060892 4.2 video length: 3:10 379319375 4.2 video length: 2:09 379150092 5.9 video length: 4:50 379319558 5.9 video length: 4:29 381974697 5.4 video length: 9:02

There was not enough data to do any robust data analysis for media_views.csv, as there were only five videos in the data set. One could infer that the video with ID 381974697 was skipped over almost completely by the most students compared to the other four videos because it was by far the longest, at 9 minutes and 2 seconds long. But this is still only one data point.

Load and preprocess page_views.csv and responses.csv

```
page_views_2 <- read.csv("C:/Users/maxch/Downloads/Data/Random Sample of Data Files/page_views.csv")
page_views <- page_views_2
page_views <- page_views[!is.na(page_views["chapter_number"]),
]
```

```
# combine chapter and section
# numbers as done previously
page_views["chapter_section"] <- as.double(unlist(page_views["chapter_number"])) +
  as.double(unlist(page_views["section_number"]))/10
head(page_views)
```

```
##                               book release
## 1 College / Statistics and Data Science (ABC)    v5.0
## 2 College / Statistics and Data Science (ABC)    v5.0
## 3 College / Statistics and Data Science (ABC)    v5.0
## 4 College / Statistics and Data Science (ABC)    v5.0
## 5 College / Statistics and Data Science (ABC)    v5.0
## 6 College / Statistics and Data Science (ABC)    v5.0
##                               chapter
## 1 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 2 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 3 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 4 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 5 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 6 Chapter 1 - Welcome to Statistics: A Modeling Approach
##                               page chapter_number section_number
## 1      1.3 Doing Statistics with R                1              3
## 2      1.3 Doing Statistics with R                1              3
## 3      1.3 Doing Statistics with R                1              3
## 4 1.4 Introduction to R Functions                 1              4
## 5 1.4 Introduction to R Functions                 1              4
## 6 1.4 Introduction to R Functions                 1              4
##                               institution_id          class_id
## 1 04157183-8665-400a-925d-3bbb70ffe45e afcb6b4e-a0c0-46ce-b38c-c96329c91471
```

```
## 2 04157183-8665-400a-925d-3bbb70ffe45e afcb6b4e-a0c0-46ce-b38c-c96329c91471
## 3 04157183-8665-400a-925d-3bbb70ffe45e afcb6b4e-a0c0-46ce-b38c-c96329c91471
## 4 04157183-8665-400a-925d-3bbb70ffe45e afcb6b4e-a0c0-46ce-b38c-c96329c91471
## 5 04157183-8665-400a-925d-3bbb70ffe45e afcb6b4e-a0c0-46ce-b38c-c96329c91471
## 6 04157183-8665-400a-925d-3bbb70ffe45e afcb6b4e-a0c0-46ce-b38c-c96329c91471
##
##      student_id      dt_accessed was_complete
## 1 953768c5-32fa-4190-8fee-1da82d8710bc 2023-01-06T00:47:06Z      FALSE
## 2 953768c5-32fa-4190-8fee-1da82d8710bc 2023-01-14T04:41:55Z      FALSE
## 3 953768c5-32fa-4190-8fee-1da82d8710bc 2023-01-14T20:15:03Z      TRUE
## 4 953768c5-32fa-4190-8fee-1da82d8710bc 2023-01-06T00:46:58Z      FALSE
## 5 953768c5-32fa-4190-8fee-1da82d8710bc 2023-01-14T23:51:24Z      TRUE
## 6 953768c5-32fa-4190-8fee-1da82d8710bc 2023-01-14T20:28:21Z      FALSE
##      tried_again_dt tried_again_clicks engaged idle_brief idle_long off_page_brief
## 1      <NA>          0      10857          0          0          0
## 2      <NA>          0     620302      188748          0          0
## 3      <NA>          0     793964          0          0          0
## 4      <NA>          0       6250          0          0          0
## 5      <NA>          0     30885          0          0          0
## 6      <NA>          0    1413760          0          0          0
##      off_page_long review_flag chapter_section
## 1          0      FALSE          1.3
## 2          0      FALSE          1.3
## 3          0      FALSE          1.3
## 4          0      FALSE          1.4
## 5          0      FALSE          1.4
## 6          0      FALSE          1.4
```

```
responses <- read.csv("C:/Users/maxch/Downloads/Data/Random Sample of Data Files/responses.csv")
responses$chapter_section <- as.double(unlist(responses["chapter_number"])) +
  as.double(unlist(responses["section_number"]))/10
head(responses)
```

```
##
##      book      release
## 1 College / Advanced Statistics and Data Science (ABCD) v5.0-exp2
## 2 College / Advanced Statistics and Data Science (ABCD) v5.0-exp2
## 3 College / Advanced Statistics and Data Science (ABCD) v5.0-exp2
## 4 College / Advanced Statistics and Data Science (ABCD) v5.0-exp2
## 5 College / Advanced Statistics and Data Science (ABCD) v5.0-exp2
## 6 College / Advanced Statistics and Data Science (ABCD) v5.0-exp2
##
##      chapter
## 1 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 2 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 3 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 4 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 5 Chapter 1 - Welcome to Statistics: A Modeling Approach
## 6 Chapter 1 - Welcome to Statistics: A Modeling Approach
##
##      page chapter_number section_number review_flag
## 1 1.3 Doing Statistics with R          1          3      FALSE
## 2 1.3 Doing Statistics with R          1          3      FALSE
## 3 1.3 Doing Statistics with R          1          3      FALSE
## 4 1.3 Doing Statistics with R          1          3      FALSE
## 5 1.3 Doing Statistics with R          1          3      FALSE
## 6 1.3 Doing Statistics with R          1          3      FALSE
##
##      institution_id      class_id
## 1 292cff87-3c74-4e94-8622-233afb0427dd 6fbf5a0a-cf5d-4567-89b5-eb5c4a16c4ab
```

```

## 2 292cff87-3c74-4e94-8622-233afb0427dd 6fbf5a0a-cf5d-4567-89b5-eb5c4a16c4ab
## 3 292cff87-3c74-4e94-8622-233afb0427dd 6fbf5a0a-cf5d-4567-89b5-eb5c4a16c4ab
## 4 292cff87-3c74-4e94-8622-233afb0427dd 6fbf5a0a-cf5d-4567-89b5-eb5c4a16c4ab
## 5 292cff87-3c74-4e94-8622-233afb0427dd 6fbf5a0a-cf5d-4567-89b5-eb5c4a16c4ab
## 6 292cff87-3c74-4e94-8622-233afb0427dd 6fbf5a0a-cf5d-4567-89b5-eb5c4a16c4ab
##
##          student_id item_id item_type
## 1 0780b2c5-b260-4a31-816e-e91ce8b403a0    ch1-0    code
## 2 ceacd30b-36ff-413e-9d7d-c389ad5086cf    ch1-0    code
## 3 ba593149-8b3c-4e1e-8df9-2b470c49a4f4    ch1-0    code
## 4 1bacdb33-5165-44c1-9aa5-b6a904e6c463    ch1-0    code
## 5 646acd2f-5a8b-4629-b753-faff3ff49b8c    ch1-0    code
## 6 1bf8f1f2-fbdd-4e08-9f12-daa182d154c3    ch1-0    code
##
##                                     response
## 1                                     # type whatever you want\n# see... blah blah blah\n
## 2                                     # type whatever you want\n# see... blah blah blah\n# happy birthday to you\n
## 3                                     # type whatever you want\n# see... blah blah blah\n
## 4                                     # type whatever you want\n# see... blah blah blah blah\n
## 5                                     # type whatever you want\n# see... blah blah blah\n# typing whatever I want!
## 6 # I love the way this text book is written\n# \\" You can't see me \\" - comments to R\n
##
##          prompt points_possible
## 1 # type whatever you want\n# see... blah blah blah\n          1
## 2 # type whatever you want\n# see... blah blah blah\n          1
## 3 # type whatever you want\n# see... blah blah blah\n          1
## 4 # type whatever you want\n# see... blah blah blah\n          1
## 5 # type whatever you want\n# see... blah blah blah\n          1
## 6 # type whatever you want\n# see... blah blah blah\n          1
##
## points_earned      dt_submitted completes_page attempt user_agent
## 1          1 2023-04-04T06:19:03Z          TRUE          1      <NA>
## 2          1 2023-04-04T05:36:51Z          TRUE          1      <NA>
## 3          1 2023-04-09T23:27:47Z          FALSE          1      <NA>
## 4          1 2023-04-09T22:55:57Z          FALSE          1      <NA>
## 5          1 2023-04-09T23:10:08Z          TRUE          1      <NA>
## 6          1 2023-04-09T23:58:49Z          FALSE          1      <NA>
##
## lrn_response_id lrn_activity_reference lrn_question_reference
## 1          <NA>          <NA>          <NA>
## 2          <NA>          <NA>          <NA>
## 3          <NA>          <NA>          <NA>
## 4          <NA>          <NA>          <NA>
## 5          <NA>          <NA>          <NA>
## 6          <NA>          <NA>          <NA>
##
## lrn_question_position lrn_type lrn_dt_started lrn_dt_saved lrn_status
## 1          NA      <NA>          <NA>          <NA>      <NA>
## 2          NA      <NA>          <NA>          <NA>      <NA>
## 3          NA      <NA>          <NA>          <NA>      <NA>
## 4          NA      <NA>          <NA>          <NA>      <NA>
## 5          NA      <NA>          <NA>          <NA>      <NA>
## 6          NA      <NA>          <NA>          <NA>      <NA>
##
## lrn_option_0 lrn_option_1 lrn_option_2 lrn_option_3 lrn_option_4 lrn_option_5
## 1          <NA>          <NA>          <NA>          <NA>          <NA>      <NA>
## 2          <NA>          <NA>          <NA>          <NA>          <NA>      <NA>
## 3          <NA>          <NA>          <NA>          <NA>          <NA>      <NA>
## 4          <NA>          <NA>          <NA>          <NA>          <NA>      <NA>
## 5          <NA>          <NA>          <NA>          <NA>          <NA>      <NA>
## 6          <NA>          <NA>          <NA>          <NA>          <NA>      <NA>

```

```
##   lrn_option_6 lrn_option_7 lrn_option_8 lrn_option_9 lrn_option_10
## 1          <NA>          <NA>          <NA>          <NA>          <NA>
## 2          <NA>          <NA>          <NA>          <NA>          <NA>
## 3          <NA>          <NA>          <NA>          <NA>          <NA>
## 4          <NA>          <NA>          <NA>          <NA>          <NA>
## 5          <NA>          <NA>          <NA>          <NA>          <NA>
## 6          <NA>          <NA>          <NA>          <NA>          <NA>
##   lrn_option_11 chapter_section
## 1          <NA>             1.3
## 2          <NA>             1.3
## 3          <NA>             1.3
## 4          <NA>             1.3
## 5          <NA>             1.3
## 6          <NA>             1.3
```

Analysis of page_views.csv and responses.csv

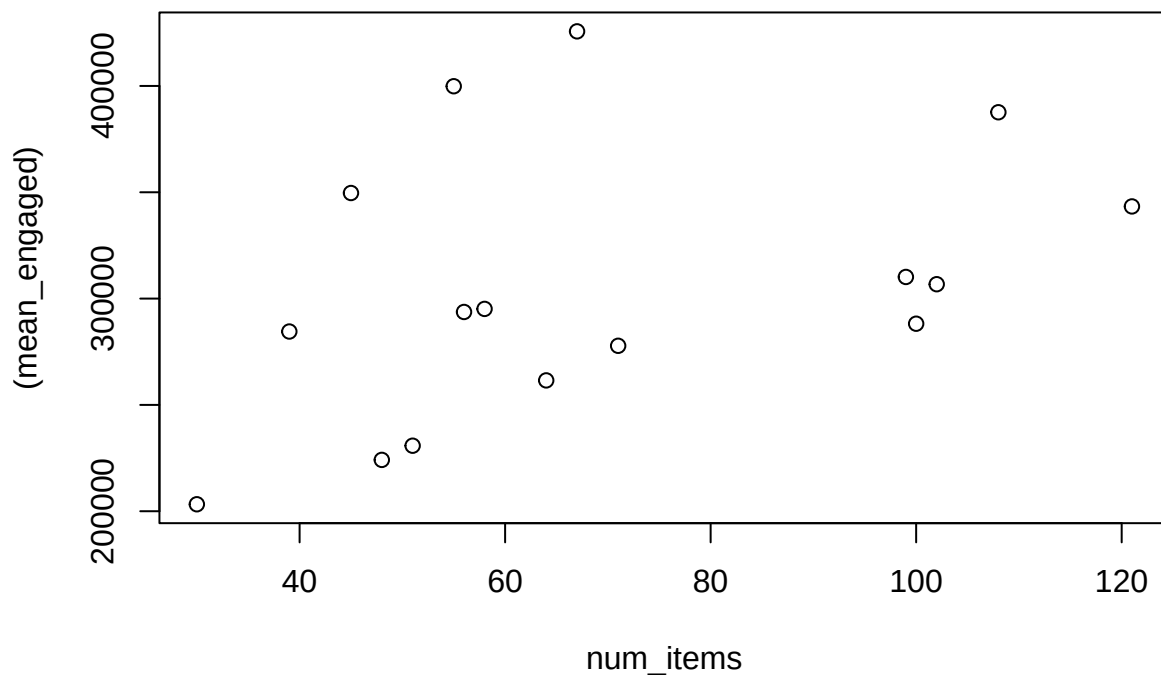
```
info_by_ch <- page_views %>%
  group_by(chapter_number) %>%
  summarize(mean_engaged = mean(engaged,
    na.rm = TRUE))
items_by_ch <- responses %>%
  group_by(chapter_number) %>%
  summarize(num_items = length(unique(item_id)))
items_by_ch <- items_by_ch[-(nrow(items_by_ch)),
  ]
info_by_ch <- cbind(info_by_ch,
  items_by_ch[, 2])
info_by_ch
```

```
##   chapter_number mean_engaged num_items
## 1              1    284508.3         39
## 2              2    310171.3         99
## 3              3    387618.8        108
## 4              4    343333.3        121
## 5              5    277795.5         71
## 6              6    306755.4        102
## 7              7    230814.6         51
## 8              8    295148.2         58
## 9              9    288180.4        100
## 10             10    349638.9         45
## 11             11    399859.4         55
## 12             12    425660.4         67
## 13             13    261521.7         64
## 14             14    293701.3         56
## 15             15    224143.0         48
## 16             16    203274.0         30
```

```
engagement_num_items_by_ch <- with(info_by_ch,
  summary(lm((mean_engaged) ~
    (num_items))))
engagement_num_items_by_ch
```

```
##
## Call:
```

```
## lm(formula = (mean_engaged) ~ (num_items))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -66217 -39982 -14697  17184 122889
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 242506.9    41889.0   5.789 4.69e-05 ***
## num_items      899.5       561.6   1.602   0.132
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 60090 on 14 degrees of freedom
## Multiple R-squared:  0.1548, Adjusted R-squared:  0.09448
## F-statistic: 2.565 on 1 and 14 DF,  p-value: 0.1316
with(info_by_ch, plot((mean_engaged) ~
  (num_items))))
```



Plotting the number of items per chapter versus the mean engagement per chapter does not yield a significant correlation, as indicated by the t-test above. However, a relationship could still be found if the data were split by section rather than by chapter. Over an entire chapter, the engagement could average out between all its sections to be fairly similar to all the other chapters, concealing any variability between them.

```
info_by_section <- page_views %>%
  group_by(chapter_section) %>%
```



```

    summarize(mean_engaged = mean(engaged,
      na.rm = TRUE))
items_by_section <- responses %>%
  group_by(chapter_section) %>%
  summarize(num_items = length(unique(item_id)))
items_by_section <- items_by_section[!is.na(items_by_section[,
  1]), ]
info_by_section <- cbind(info_by_section,
  items_by_section[, 2])
head(info_by_section)

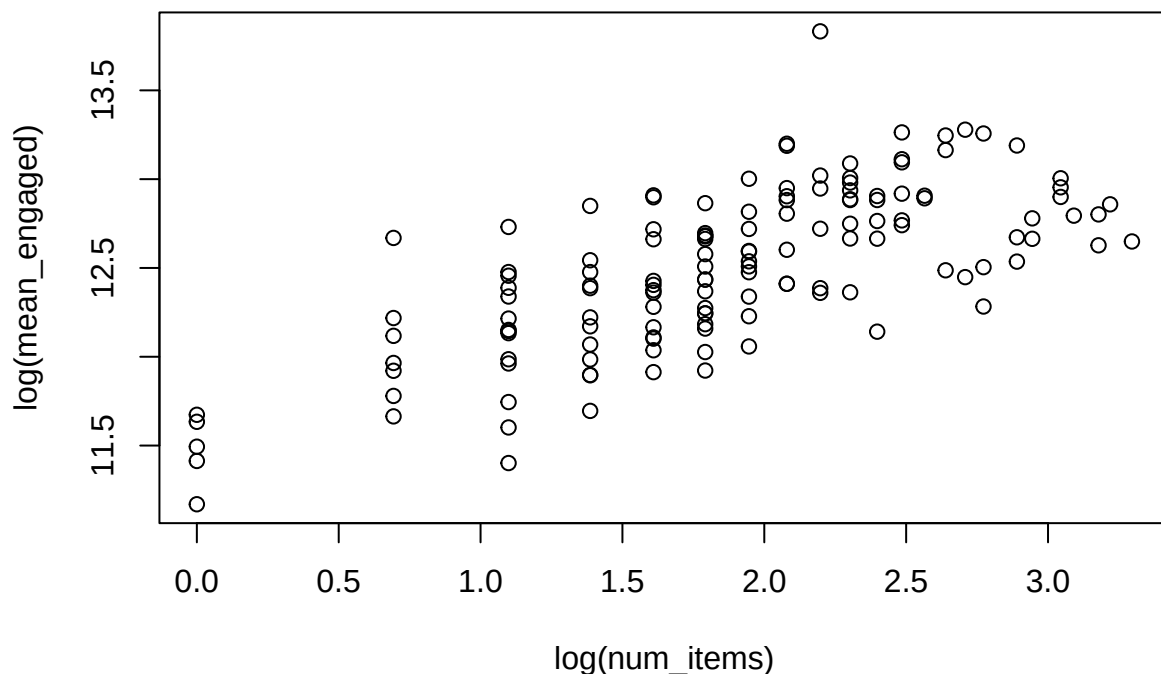
##   chapter_section mean_engaged num_items
## 1             1.3    160296.75         4
## 2             1.4    150595.22         6
## 3             1.5    584529.31        15
## 4             1.6     98081.73          1
## 5             1.7    264797.82        14
## 6             2.1    174365.17          4

engagement_num_items_by_section <- with(info_by_section,
  summary(lm(log(mean_engaged) ~
    log(num_items))))
engagement_num_items_by_section

##
## Call:
## lm(formula = log(mean_engaged) ~ log(num_items))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.75296 -0.24820 -0.00914  0.22989  1.18395
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   11.66069    0.07625   152.93  <2e-16 ***
## log(num_items)  0.44945    0.03821    11.76  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3214 on 137 degrees of freedom
## Multiple R-squared:  0.5024, Adjusted R-squared:  0.4988
## F-statistic: 138.3 on 1 and 137 DF, p-value: < 2.2e-16

with(info_by_section, plot((log(mean_engaged) ~
  log(num_items))))

```



Using a log-transformation on both variables, the relationship between the number of items per section and the mean engagement per section is clearly significant according to the above t-test. This makes sense, because engagement is really a measure for the total time spent being active on the section's pages, and if there are more items in a section, then students would naturally spend more time on the section going through each item.

Loading and preprocessing checkpoints_eoc.csv (end-of-chapter checkpoints)

```
cp_eoc <- read.csv("C:/Users/maxch/Downloads/Data/Random Sample of Data Files/checkpoints_eoc.csv")
eoc_by_ch <- cp_eoc %>%
  group_by(chapter_number) %>%
  summarize(mean_eoc = mean(EOC,
    na.rm = TRUE))
cbind(info_by_ch, eoc_by_ch[, 2])
```

##	chapter_number	mean_engaged	num_items	mean_eoc
## 1	1	284508.3	39	0.7949015
## 2	2	310171.3	99	0.6869427
## 3	3	387618.8	108	0.6393878
## 4	4	343333.3	121	0.6087486
## 5	5	277795.5	71	0.6360003
## 6	6	306755.4	102	0.6007507
## 7	7	230814.6	51	0.5888460
## 8	8	295148.2	58	0.6003845
## 9	9	288180.4	100	0.5823940
## 10	10	349638.9	45	0.5616277

```
## 11          11      399859.4      55 0.4854980
## 12          12      425660.4      67 0.5295909
## 13          13      261521.7      64 0.5164202
## 14          14      293701.3      56 0.4713821
## 15          15      224143.0      48 0.4900817
## 16          16      203274.0      30 0.3879706
```

```
summary(lm(eoc_by_ch$mean_eoc ~
  info_by_ch$num_items))
```

```
##
## Call:
## lm(formula = eoc_by_ch$mean_eoc ~ info_by_ch$num_items)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.13706 -0.05212 -0.02060  0.03870  0.25879
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.4881042  0.0647191   7.542  2.7e-06 ***
## info_by_ch$num_items 0.0012309  0.0008677   1.419    0.178
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09285 on 14 degrees of freedom
## Multiple R-squared:  0.1257, Adjusted R-squared:  0.06323
## F-statistic: 2.012 on 1 and 14 DF,  p-value: 0.1779
```

```
plot((eoc_by_ch$mean_eoc ~ info_by_ch$mean_engaged))
```

